

BASML Cotton & Yarn Intelligence Platform

Comprehensive Operations, Methodology, & Data Analytics Audit Report

Report Generated: May 30, 2026
Deployment Target: GitHub Pages Live System
Audited Version: v1.1.0-Stabilized

1. Executive Summary & Platform Overview

The **BASML Cotton & Yarn Intelligence Platform** is a corporate-grade Commodity Decision Support System (CDSS) built to manage raw material volatility, optimize sourcing channels, and enforce strict yarn quality standards for mills in Tamil Nadu. Raw cotton represents up to 60-65% of yarn manufacturing expenses. In an environment marked by weather irregularities (such as El Niño), international trade disputes, and currency fluctuations, the platform integrates live financial feeds, regulatory floor prices, and manufacturing conversion metrics into a unified cockpit.

The platform is structured around a multi-tab sidebar navigation, feeding live metrics into analytical dashboards:

- **Global Focus:** Monitors international price benchmarks (ICE, Cotlook A-Index) and global supply/demand balances.
- **India Focus:** Tracks domestic spot prices, APMC mandi arrivals, and central buffer stock levels.
- **Cotton Markets:** Analyzes ginning out-turn parity (GOT%) and pressed candy profit margins.
- **Yarn Markets:** Compares yarn count prices and maps spinning margin spreads relative to raw cotton.
- **Import & Export Plan:** Interactive logistics calculators evaluating global import landed costs and export realizations.
- **Analysis Suite:** Houses the 80+ Variety Explorer, 12-Month Crop Calendar, Technical Price Indicators, and MSP floor reference sheets.
- **Yarn Quality (Global Focus):** Real-time Statistical Process Control (SPC) monitor displaying count deviations and defect indexes.

2. Global Focus (Tab 1)

Purpose & Business Reason

The Global Focus view provides international market awareness. Indian spinners rely on imported long-staple cotton (e.g., US Pima, Egyptian Giza) to spin premium compact counts. Therefore, tracking the **ICE US Cotton No. 2 Futures** and currency conversion rates is vital for identifying favorable import arbitrage windows and hedging currency exposure.

Calculations & Formulas

Formula 1: ICE US Cotton INR Candy Parity Equivalent

$$ICE\ Parity\ Candy\ (INR) = ICE\ Price\ (cents/lb) \times 7.84 \times USD/INR\ Exchange\ Rate$$

Where: 7.84 is the conversion factor translating standard US pounds to one Indian Candy (784 lbs = 355.62 kg).

How to Read the Visual Elements

- **Tickers & Cards:** Displays the current price of ICE Futures, Cotlook A-Index, Brazilian Index, Chinese Index, and Pakistani Spot Rates.
- **Global Crop Balance Sheet:** Lists years against global production, consumption, and ending stock indexes, alerting buyers to global inventory crunches.



Figure 2.1: Global Focus Dashboard showing international price feeds and USDA balance sheet data.

3. India Focus (Tab 2)

Purpose & Business Reason

The India Focus dashboard acts as the primary cockpit for domestic sourcing. Since domestic varieties like Shankar-6 and MCU-5 form the baseline feed for standard carded/combed counts, monitoring Mandi arrivals, Cotton Corporation of India (CCI) procurement, and spot price movements ensures that procurement teams buy when the physical arrivals peak and prices dip.

Calculations & Formulas

Formula 2: MSP-Aware Shankar-6 Spot Pricing Parity

$$\text{Shankar-6 Spot Price} = \max\left(54500, \left\lfloor \text{ICE INR Equivalent} \times 0.965 \right\rfloor\right)$$

Where: ₹54,500 is the regulatory Minimum Support Price (MSP) floor candy equivalent. A discount factor of 0.965 accounts for higher average trash content in domestic lint.

How to Read the Visual Elements

- **Domestic Price list:** Highlights regional variety rates per candy and bale.
- **Mandi Arrivals Bar Chart:** Plots weekly raw arrivals in regional markets, helping predict near-term spot price declines when arrival curves spike.

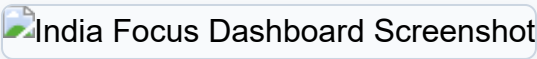


Figure 3.1: India Focus Dashboard tracking Shankar-6, MCU-5, and regional Mandi arrivals.

4. Cotton Markets & Ginning Parity (Tab 3)

Purpose & Business Reason

Ginning Out-Turn (GOT%) represents the percentage of lint extracted from raw seed cotton (Kapas). Different regions and crop varieties produce different GOT yields. Sourcing cotton with a high GOT percentage maximizes processed fiber yield and directly increases the ginning margin spread.

Calculations & Formulas

Formula 3: Ginned Candy Cost & Parity Spread

$$\text{Clean Cotton Cost} = \frac{\text{Mandi Candy Price} \times (1 - \text{Waste}\%)}{\text{Total Ginned Candy Cost}}$$
$$\text{Net Ginning Margin (Spread)} = \text{Market Selling Price} - \text{Total Ginned Candy Cost}$$

How to Read the Visual Elements

- Interactive Parity Calculator:** Calculates the raw Kapas value per quintal and total candy cost based on the selected variety, showing the exact net ginning margin.
- Indian State-Wise GOT Parity Table:** Lists production alongside regional average yields and Ginning Out-Turn percentages.

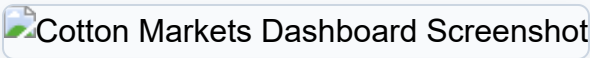


Figure 4.1: Cotton Markets Dashboard displaying crop parity calculators and state-wise ginning parameters.

5. Yarn Markets & Spinning Parity (Tab 4)

Purpose & Business Reason

Spinning mills face severe margin squeezes. When cotton prices rise, yarn prices do not always rise in tandem, resulting in margin contraction. The Yarn Markets tab helps spinning mills monitor count-specific **Spinning Spreads** (Yarn Price minus Raw Cotton Cost & conversion overheads) to determine which counts are profitable to spin.

Calculations & Formulas

Formula 4: Yarn Count Physical Costing & Margin Spreads

$$\text{Clean Cotton Cost (\text{₹}/\text{kg})} = \frac{\text{Shankar-6 Spot Price (\text{₹}/\text{Candy})}}{355.62 \times \text{Yield}}$$
$$\text{Total Yarn Manufacturing Cost} = \text{Clean Cotton Cost} + \text{Mill Conversion Cost}$$
$$\text{Spinning Margin Spread (\text{₹}/\text{kg})} = \text{Yarn Market Selling Price} - \text{Total Yarn Manufacturing Cost}$$

Where Yield and Conversion Costs vary by count spinning system:

- 20s Carded: Yield = 91%, Conversion Cost = ₹45/kg
- 30s Combed: Yield = 80%, Conversion Cost = ₹65/kg
- 40s Compact: Yield = 77%, Conversion Cost = ₹95/kg
- 80s Compact: Yield = 50%, Conversion Cost = ₹265/kg

How to Read the Visual Elements

- **Split Tables:** Displays pure cotton yarns and synthetic/blends tables side-by-side.
- **Interactive Spinning Margin Calculator:** Computes the count-specific yarn price, clean raw material equivalent, fixed power/labor conversion cost, and finished manufacturing cost to evaluate profit spreads.



Figure 5.1: Yarn Markets Dashboard showing split count lists and the Spinning Parity Calculator.

6. Import & Export Sourcing Desk (Tab 5)

Purpose & Business Reason

Spinning mills must evaluate whether importing foreign cotton is more cost-effective than purchasing domestic out-of-state cotton (e.g., from Gujarat). Similarly, export realizations must be evaluated against local sales. The Import & Export Desk provides full logistics visibility, factoring in customs, surcharges, and local port freight.

Calculations & Formulas

Formula 5: Sourcing Landed Cost Equations

$$\begin{aligned} \text{CIF (INR)} &= \text{FOB Price (INR)} + \text{Ocean Freight} + \text{Marine Insurance} \\ \text{Basic Customs Duty (BCD)} &= \text{CIF} \times 10\% \\ \text{Agri Infrastructure Cess (AIDC)} &= \text{CIF} \times 5\% \\ \text{Social Welfare Surcharge (SWS)} &= (\text{BCD} + \text{AIDC}) \times 10\% \\ \text{Global Landed Cost} &= \text{CIF} + \text{BCD} + \text{AIDC} + \text{SWS} + \text{Port Handling} + \text{Inland Road Freight} \\ \text{Domestic Interstate Landed Cost} &= \text{Mandi Spot Price} + \text{Inland Freight} + \text{GST (RCM 5\%)} + \text{Mandi Cess (1\%)} + \text{Handling} \end{aligned}$$

How to Read the Visual Elements

- Sourcing Landed Cost Calculator:** Computes global container freight and duties vs domestic road transportation costs.
- Yarn Export Net Realization:** Deducts port commissions, export wrapping, and sea logistics from destination sale values to show net mill-gate profits.



Figure 6.1: Import & Export Planning Desk displaying Landed Cost Calculator fields.

7. Live News & Arbitrage Playbook (Tab 6)

Purpose & Business Reason

Textile markets respond heavily to news events, macro policies, and agricultural warnings. The Playbook establishes operational SOPs (Standard Operating Procedures) for buyers during weather anomalies (such as El Niño droughts) or tariff changes. This helps the mill stay proactive rather than reactive during supply shocks.

How to Read the Visual Elements

- **Arbitrage Rules:** Guides buyers on when to buy long-term forwards (e.g. when ICE price dips below MSP margins).
- **Macro Incident Grid:** Connects global news (e.g., Canal transit constraints) to affected segments and finished yarn price implications.



Figure 7.1: Live News, Incidents, & Sourcing Playbook Desk.

8. Variety Explorer Suite (Analysis Sub-Tabs)

8.1. 80+ Variety Explorer

Purpose: A searchable, categorized library of domestic cottons, global benchmarks, and yarn varieties. It permits buyers to cross-examine properties (staple, micronaire, price) and compare up to three varieties in a spec matrix.

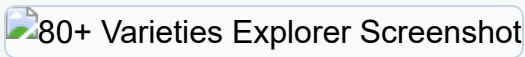


Figure 8.1: Variety Explorer sub-tab featuring the Comparison Matrix.

8.2. Crop Cycle & Pricing Calendar

Purpose: Displays sowing, growth, harvesting cycles, and seasonal spot price expectations across 12 months. It guides buyers to build stockpiles during low-price harvest windows (Nov-Feb) and minimize buying during high-price sowing cycles (July-Sept).



Figure 8.2: 12-Month Sowing & Pricing Seasonality Calendar.

8.3. Technical Price Analytics & Crossover Alerts

Purpose: Evaluates active spot price trend charts and moving averages (10-day MA, 50-day MA). It computes support/resistance thresholds and volatility indices, generating buy/sell momentum signals (Golden Cross / Death Cross).

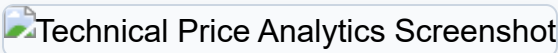


Figure 8.3: Technical price analytics with 60-day moving average chart and crossover alerts.

8.4. State-Wise MSP Floor Reference Sheets

Purpose: Monitors regulatory floor prices (Minimum Support Prices) across Indian states, helping procurement managers assess state-wise crop margins.



Figure 8.4: Government MSP pricing table showing year-on-year increases.

9. Yarn Quality & Statistical Process Control (Tab 7)

Purpose & Business Reason

Spinning floor quality stability dictates export contract compliance. Count variations cause fabric shading defects (barre), and weak strands trigger high-speed loom warp breaks. By monitoring Uster benchmarks and count runs, spinning technicians can stop machinery immediately upon observing count drift, avoiding waste.

Calculations & Formulas

Formula 6: SPC Count Limits & CV% Formulas

$$\text{Upper Control Limit (UCL)} = \text{Target Count} + \left(\text{Target Count} \times 2.0\% \right)$$
$$\text{Lower Control Limit (LCL)} = \text{Target Count} - \left(\text{Target Count} \times 2.0\% \right)$$
$$\text{Count Coefficient of Variation (CV\%)} = \frac{\text{Standard Deviation of Yarn Count}}{\text{Average Measured Count}} \times 100$$

Where: A CV% below 1.0% represents top-tier global spinning consistency.

How to Read the Visual Elements

- SPC Run Chart:** Plots 15 daily batch runs against the Center Line (Target Count), UCL, and LCL. Points outside the boundaries indicate drafting zone instability.
- IPI Defect Bar Chart:** Plots Thin, Thick, and Nep defects per kilometer against the top Uster 5% world benchmark.



Figure 9.1: Yarn Quality Dashboard showing the Count Run SPC chart and IPI defect breakdown.